E-tiGo — Open Questions Decision Report

Phase 1 Backend — Complete Decision Record for All Outstanding Questions

Prepared by: Engineering Team | **Date:** 13 October 2026 | **Document version:** 3.0 | **Status:** All Phase 1 Decisions Resolved & Implemented

Executive Summary

This document records the final decisions for **all 26 Phase 1 open questions** identified during E-tiGo backend development. Every question has been resolved and the corresponding code has been implemented and verified.

Questions are grouped into six categories: Provider & Integration (5), Business Rules (6), Detection & Enforcement (2), EV Charging (2), Wallet & Ledger (9), and App Configuration (2).

Items tagged NEW were decided and implemented in the October 2026 update cycle. All other decisions were resolved during earlier development phases.

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A. Provider & Integration Decisions

OQ-01 — SMS/OTP Gateway

Question

Which SMS/OTP gateway should E-tiGo use for phone number verification?

Decision

WhatsApp Business API is the primary OTP delivery channel.

- `WhatsAppGateway` — auto-selected in `AppServiceProvider` when `WHATSAPP_PHONE_NUMBER_ID` and `WHATSAPP_ACCESS_TOKEN` env vars are set
- `LogSmsGateway` — dev/testing fallback (logs OTP to Laravel log file)
- WhatsApp was chosen over traditional SMS for cost efficiency and higher delivery rates in the Nigerian market

OQ-02 — Maps & Geocoding Provider

Question

Which maps/geocoding provider should E-tiGo use for distance, directions, and place search?

Decision

Mapbox for directions, distance, and duration calculations. Multi-provider failover for place search.

- `MapboxGateway` — primary directions provider; `HaversineMapsGateway` as offline fallback
- `FailoverPlaceSearchGateway` — configurable provider chain: Mapbox ? Geoapify ? LocationIQ ? Photon
- Provider order configurable via `PLACE_SEARCH_DRIVERS` env var
- All providers auto-selected based on available API keys in `config/services.php`

OQ-03 — Card Payment Gateway

Question

Which card payment gateway should E-tiGo use for card tokenization and wallet operations?

Decision

Paystack — Nigeria's leading payment processor.

- `PaystackPaymentGateway` — card tokenization, charge authorization, and refunds
- `PaystackWalletGateway` — wallet top-ups and transfers
- `FakePaymentGateway` — dev/testing fallback (activated when `PAYSTACK_FAKE=true` or keys absent in local/staging)
- Config: `PAYSTACK_SECRET_KEY` / `PAYSTACK_PUBLIC_KEY` in `config/services.php`

OQ-04 — Emergency Services Dispatch NEW

Question

Which emergency-services dispatch integration should E-tiGo use for its SOS/panic button feature?

Industry Benchmark

In Nigeria, dedicated emergency dispatch APIs (such as RapidSOS) are unavailable or prohibitively expensive. Bolt Nigeria, Uber Nigeria, and InDrive route panic events through internal safety teams. Nigeria's 112 emergency line is the standard public safety number.

Decision

> **No external dispatch API.** The SOS panic button will:
>
> - Send GPS + trip details via **WhatsApp Business API** to an internal E-tiGo safety number
> - Auto-dial **112** (Nigeria's emergency line) via the `auto_dial_emergency` response field
> - **Notify Admin dashboard in real-time** via WebSocket broadcast on initial trigger
> - Store each incident with encrypted telemetry for audit trail

> **Implementation:** `SosIncidentTriggered` event (broadcasts to `admin.sos`), `SendSosWhatsAppAlertJob`, `auto_dial_emergency: "112"` in API response, `WHATSAPP_SAFETY_NUMBER` config.

OQ-17 — Minimum App Version Enforcement NEW

Question

What are the minimum supported iOS and Android versions, and how is version enforcement handled?

Decision

> **Minimum version set to launch version (1.0.0 for both iOS and Android).**
>
> - `min_app_version_ios` / `min_app_version_android` stored in AppSettings — Admin-adjustable at runtime
> - Returned via the `GET /api/v1/config` public endpoint (no auth required)
> - `force_update` flag available for critical updates that block app usage entirely

> **Implementation:** `ConfigController` — `GET /api/v1/config` returns min versions, force_update, emergency_number, support contacts.

B. Business Rule Decisions

OQ-05 — Ride Cancellation Fee Policy NEW

Question

Should Phase 1 include a cancellation fee, and under what conditions?

Decision

No cancellation fee in Phase 1. Riders may cancel freely at any point. The `ride.free_cancellation_window_seconds` config is retained for future use.

Phase 2 recommendation: Introduce ?300–?500 flat fee after 60-second grace window, with automatic waiver if driver GPS shows no movement toward pickup within 2 minutes.

OQ-06 — Driver Payout Frequency NEW

Question

How often should driver earnings be paid out — on-demand only, or with an automated schedule?

Decision

Both on-demand and weekly auto-payout.

- **On-demand**: driver requests via app, Admin approves, processed via Paystack Transfer API
- **Weekly auto-payout** (opt-in): every Monday at 06:00 WAT, auto-approved
- **Admin-configurable** payout thresholds via `AdminWalletSettingsController`

Implementation: `payout_preference` column on drivers, `PayoutPreference` enum, `PUT /driver/payouts/preference` endpoint, `ProcessWeeklyAutoPayoutJob` scheduled Monday 06:00 WAT.

OQ-07 — Admin Role Structure

Question

Should E-tiGo use a single Admin role or role-based tiers?

Decision

Role-based tiers implemented with five distinct roles:

- `super_admin` — full system access
- `operations` — ride management, driver oversight
- `safety_operator` — SOS incidents, flagged trips
- `support` — disputes, passenger issues
- `finance` — payouts, reconciliation

Enforced via `EnsureAdminRole` middleware with route-level role requirements.

OQ-08 — Admin Refund Authority

Question

Can Admin issue refunds and fare adjustments when resolving disputes?

Decision

> **Yes.** `AdminRefundController@refundToWallet` — `POST /admin/rides/{ride}/refund`
> - Full or partial refund to passenger wallet via `WalletRefundService`
> - Linked to dispute if one exists
> - Claws back driver earnings via `WalletRefundService@clawbackDriverEarnings()`

OQ-09 — Gamification Tier Naming

Question

Which tier naming convention: Option A (Bronze ? Diamond) or Option B (Seed ? Forest)?

Decision

> **Option A** — industry-standard naming:
> - Bronze (0 pts) ? Silver (500) ? Gold (2,000) ? Platinum (5,000) ? Diamond (15,000)
> - Implemented in `TierLevel` enum and seeded via `TierConfigSeeder`

OQ-10 — Carbon Emissions Baseline

Question

What baseline and per-vehicle emissions data should the carbon scoring engine use?

Decision

> **EU/IPCC averages adjusted for ride-sharing occupancy**, configured in `config/gamification.php`:
> - Baseline: 120 g CO?/km (average private car)
> - Economy: 65 g/km | Comfort: 80 g/km | Premium: 95 g/km | EV: 0 g/km | Default: 75 g/km

OQ-11 — Surge Pricing Scope

Question

Is surge/dynamic pricing included in Phase 1?

Decision

> **Yes — fully built.**
> - `SurgePricingService` evaluates active rules by priority: manual ? time_based ? demand_based
> - `SurgeRule` model with `AdminSurgeRuleController` CRUD + toggle
> - Integrated into `FareEstimationService`; multiplier range 1.00–5.00

OQ-12 — Promo Code Application

Question

Should promos be auto-applied or code-entry only in Phase 1?

Decision

Code-entry only in Phase 1.

- `PromoController@validate` — validates user-entered codes against eligibility rules
- `PromoController@available` — lists promos eligible for the authenticated user
- Auto-apply deferred to Phase 2

C. Detection & Enforcement Thresholds

OQ-13 — GPS Collocation & Jitter Tolerance

Question

What GPS collocation threshold and jitter tolerance should the anti-offline-trip detection engine use?

Decision

All thresholds configurable via `config/offline_detection.php` and env vars:

- Collocation threshold: **200m**
- Route match minimum: **60%**
- Minimum GPS data points: **3**
- Pickup proximity: **300m**
- GPS sample interval: **30 seconds**
- Monitoring window: **15 minutes**

OQ-14 — False-Positive Rate & L3 Overturn Authority

Question

What is the false-positive rate target and who can overturn L3 permanent deactivation?

Decision

Super admin can overturn L3 deactivation.

- `AdminOfflineFlagController@review` with `outcome=overturned`
- `SanctionService@reverseSanction()` reinstates driver and sends notification
- 30-day lookback window for sanction tier determination

D. EV Charging Decisions

OQ-15 — EV Charging Station Data Source NEW

Question

Where does EV station data come from — third-party API, public dataset, or proprietary entry?

Decision

E-tiGo's own proprietary charging network only.

- Admin manages stations via CRUD endpoints (including delete)
- No external data feeds — only E-tiGo-owned or approved stations
- External station owners may apply to join in Phase 2

Station data fields: name, GPS location, stall count, connector types (CCS2/CHAdeMO/Type2/GBT/Tesla), power output (kW), pricing per kWh, operating hours, real-time availability.

Implementation: Migration adds `connector_types`, `power_output_kw`, `pricing_per_kwh`, `operating_hours` to `ev_charging_stations`. `DELETE /admin/ev-stations/{station}` blocks if active reservations.

OQ-16 — EV Fee Waiver Scope

Question

Does the EV reservation fee waiver apply to drivers, passengers, or both?

Decision

> **Drivers only.**
>
> - `ReservationService` checks the requesting driver's tier via `TierGateService@isEvReservationFeeWaived()`
> - Drivers are the ones making EV reservations for their vehicles

E. Wallet & Ledger Decisions

OQ-25 — Wallet Limits

Question

What are the minimum top-up, maximum balance, and daily top-up cap values?

Decision

All amounts stored and validated in kobo. Configured in `config/wallet.php`; admin-adjustable via `AdminWalletSettingsController`:

- Minimum top-up: **?500** (50,000 kobo)
- Maximum balance: **?500,000** (50,000,000 kobo)
- Daily top-up cap: **?100,000** (10,000,000 kobo)

OQ-26 — Fallback Payment Method

Question

What happens when the passenger's wallet balance is insufficient for the final fare?

Decision

Partial settlement:

- Capture whatever balance is available in the wallet
- Record the shortfall amount
- Mark payment status as `Failed` for admin review and resolution

OQ-27 — Refund Policy NEW

Question

Should refunds go to the passenger's wallet or back to the original payment method (card)?

Decision

Wallet-credit-only refunds for V1.

- All refunds issued as instant wallet credits via `WalletRefundService`
- Card refund: **admin-only escalation path** — support admins resolving disputes via `AdminDisputeController::resolve()`
- Card refund may expand in Phase 2 if required by CBN regulations

Rationale

Wallet credits are instant, cheaper to process (no Paystack refund fees), and reduce chargeback exposure. Industry standard for Bolt, InDrive in Africa.

OQ-28 — Driver Earnings Settlement Delay

Question

Should driver earnings be settled instantly, after 24 hours, or weekly?

Decision

All three options supported:

- Configurable via `config('wallet.settlement_delay')` — default: `instant`
- Admin-adjustable at runtime via Wallet Settings endpoint
- `SettleDriverEarningsJob` runs hourly to move pending ? available when delay is 24h or weekly

OQ-29 — Cash-Ride Commission Handling

Question

For cash rides, should the platform commission be debited from the driver's earnings ledger? Should negative balances be allowed?

Decision

Debit from `DriverEarningsAvailable` (driver already has the cash).

- Negative balance allowed up to configurable threshold: `max_negative_balance` = ??5,000 (?500,000 kobo)
- Warning logged when threshold exceeded
- `hasExcessiveNegativeBalance()` helper available for ride-blocking logic

OQ-30 — Minimum Payout Amount

Question

What is the minimum amount a driver can withdraw?

Decision

?1,000 (100,000 kobo) — configured in `config/wallet.php` as `min_payout`; admin-adjustable via wallet settings endpoint.

OQ-31 — Payout Schedule

Question

Should driver payouts be on-demand requests or a fixed schedule?

Decision

On-demand with Admin approval:

- Driver requests payout via `PayoutController@store`
- Admin approves via `AdminPayoutController@approve`
- Transfer processed via Paystack Transfer API

Note: OQ-06 extends this with an optional weekly auto-payout (Monday 06:00 WAT) for drivers who opt in.

OQ-32 — Tips in Wallet Flow NEW

Question

Should riders be able to tip drivers by debiting their wallet balance?

Decision

> **No wallet-debit tips in V1.**
>
> - **Card tips:** supported — charged via Paystack (`PaymentService@addTip()`)
> - **Cash tips:** handed directly to driver; logged as `tip_amount` on Payment record
> - Wallet-debit tips deferred to Phase 2

Rationale

Deducting from a closed-loop wallet for tips creates accounting ambiguity (payment vs. transfer) and potential regulatory issues under CBN guidelines.

OQ-33 — Regulatory Compliance for Closed-Loop Wallet NEW

Question

Does E-tiGo need a separate PSP or MMO license to hold customer funds in its closed-loop wallet?

Decision

> **E-tiGo operates under Paystack's CBN-licensed PSP infrastructure.**
>
> - `PaystackWalletGateway` — all customer funds held in Paystack's regulated trust accounts
> - E-tiGo functions as a sub-wallet under Paystack's license
> - Additional safeguards: ?500,000 max balance, double-entry bookkeeping, full audit trail

> **Action required:** Execute formal partnership agreement with Paystack confirming sub-wallet arrangement before public launch.

Decision Summary Matrix

#	Question	Decision	Status	Notes
OQ-01	SMS/OTP gateway	WhatsApp Business API	**Implemented**	WhatsAppGateway + LogSmsGateway fallback
OQ-02	Maps/geocoding	Mapbox + failover place search	**Implemented**	MapboxGateway + FailoverPlaceSearchGateway
OQ-03	Payment gateway	Paystack (card + wallet)	**Implemented**	PaystackPaymentGateway + PaystackWalletGateway
OQ-04	Emergency dispatch	WhatsApp + 112 + Admin alerts	**Implemented**	NEW Broadcast + WhatsApp + 112
OQ-05	Cancellation fee	No fee in Phase 1	**Aligned**	NEW Config retained for Phase 2
OQ-06	Payout frequency	On-demand + weekly auto-payout	**Implemented**	NEW Job + preference + migration
OQ-07	Admin roles	Role-based tiers (5 roles)	**Implemented**	EnsureAdminRole middleware
OQ-08	Admin refunds	Wallet refund + driver clawback	**Implemented**	AdminRefundController + WalletRefundService
OQ-09	Tier naming	Bronze ? Diamond	**Implemented**	TierLevel enum + TierConfigSeeder
OQ-10	Emissions baseline	EU/IPCC averages	**Implemented**	config/gamification.php
OQ-11	Surge pricing	Fully built in Phase 1	**Implemented**	SurgePricingService + AdminSurgeRuleController
OQ-12	Promo application	Code-entry only	**Implemented**	PromoController validate + available
OQ-13	GPS collocation	200m threshold, 60% route match	**Implemented**	config/offline_detection.php
OQ-14	L3 overturn	Super admin can overturn	**Implemented**	SanctionService@reverseSanction
OQ-15	EV station data	Proprietary network + full CRUD	**Implemented**	NEW New columns + delete
OQ-16	EV fee waiver	Drivers only	**Implemented**	TierGateService@isEvReservationFeeWaived
OQ-17	Min app versions	1.0.0, server-side enforcement	**Implemented**	NEW /api/v1/config endpoint
OQ-25	Wallet limits	?500 min / ?500k max / ?100k daily	**Implemented**	config/wallet.php, admin-adjustable
OQ-26	Fallback payment	Partial settlement + shortfall	**Implemented**	Record shortfall, mark Failed
OQ-27	Refund policy	Wallet-credit default, card admin-only	**Aligned**	NEW Both paths correctly scoped
OQ-28	Settlement delay	Instant/24h/weekly — configurable	**Implemented**	SettleDriverEarningsJob runs hourly
OQ-29	Cash commission	Debit driver, negative to ??5k	**Implemented**	max_negative_balance in config/wallet.php
OQ-30	Min payout	?1,000 (100,000 kobo)	**Implemented**	min_payout in config/wallet.php
OQ-31	Payout schedule	On-demand + admin approval	**Implemented**	PayoutController + AdminPayoutController
OQ-32	Tips in wallet	Card tips only, no wallet-debit	**Aligned**	NEW No wallet tip path exists
OQ-33	Wallet regulatory	Paystack sub-wallet / PSP license	**Aligned**	NEW Paystack is sole gateway

Action Items & Next Steps

#	Priority	Action	Owner	Timeline
1	HIGH	Execute formal sub-wallet partnership agreement with Paystack confirming regulatory coverage	Legal / BD	Before public launch

2	HIGH	Set up `WHATSAPP_SAFETY_NUMBER` and create `sos_emergency_alert` WhatsApp template in Meta Business Manager	Ops / Eng	Before public launch
3	MED	Coordinate with mobile team: (a) 112 auto-dial via `auto_dial_emergency`, (b) force-update gate via `/api/v1/config`	Mobile Team	Sprint 1 post-approval
4	LOW	Design Phase 2 "apply to charge" onboarding form for external EV station partners	Product	Phase 2 planning
5	LOW	Plan Phase 2 cancellation fee: ?300–?500 flat after 60s grace window	Product	Phase 2 planning
6	LOW	Evaluate wallet-debit tips feasibility if rider demand materialises and CBN guidance clarifies	Product / Legal	Phase 2 planning